BECE302L — CONTROL SYSTEMS

Winter Semester 2025–26

ADAPTIVE DRIVER-BEHAVIOR-AWARE SPEED CONTROL SYSTEM

Project Technical Report

MATLAB / Simulink Implementation & Full Analysis P | PI | PID (Fixed) | Adaptive Gain-Scheduled PID

State-Space | Root Locus | Bode | Nyquist | Disturbance Analysis

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BECE302L Control Systems | Winter 2025–26

Abstract

This report presents a comprehensive design, simulation, and analysis of an Adaptive Driver-Behavior-Aware Speed Control System for a vehicle modelled in MATLAB. The system models a **1200 kg vehicle** with a damping coefficient of **50 N·s/m**, targeting a reference speed of **30 m/s (108 km/h)**. Three fixed controllers — Proportional (P), Proportional-Integral (PI), and PID — are designed and compared. An adaptive gain-scheduling PID accommodates three driver profiles: Smooth, Normal, and Aggressive.

Frequency-domain stability analysis uses Bode plots and Nyquist diagrams. Root locus plots characterise closed-loop pole migration. A state-space formulation enables pole placement via full-state feedback. Disturbance rejection analysis quantifies each controller's ability to recover from a 5 m/s impulse (headwind). The key finding: **Adaptive PID with gain scheduling outperforms the fixed PID** across all metrics, delivering reduced overshoot, faster settling, and superior disturbance rejection.

1. System Modelling — Vehicle Plant

1.1 Physical Model

The vehicle is modelled using Newton's second law. The net force equals mass × acceleration; the resisting force is proportional to velocity through the damping coefficient b:

$$m \cdot dv/dt = u(t) \quad b \cdot v(t)$$

- **m** = 1200 kg (vehicle mass)
- **b** = 50 N·s/m (aerodynamic drag + friction)
- **v(t)** = vehicle speed [m/s] — output
- **u(t)** = control force input [N]

1.2 Transfer Function G(s)

Taking the Laplace transform (zero initial conditions):

$$ms \cdot V(s) + b \cdot V(s) = U(s)$$

$$G(s) = V(s) / U(s) = 1 / (ms + b) = 1 / (1200s + 50)$$

Key plant parameters:

- DC gain = 1/b = 1/50 = 0.02
- Time constant = m/b = 1200/50 = 24 s
- Open-loop pole = b/m = 0.0417 rad/s (stable, real, negative)

1.3 MATLAB Code

```
G = tf([1], [m, b]); % G(s) = 1 / (1200s + 50)
```

2. Controller Design — Formulas

2.1 Proportional (P) Controller

Produces control action proportional to the error e(t) = r(t) v(t):

$$C_P(s) = K_p = 100$$

Steady-state error (step reference):

$$e_{ss} = b / (K_p + b) = 50 / (100 + 50) \quad 0.333 \text{ (33 \% error)}$$

The P controller cannot eliminate steady-state error because it lacks an integrating term.

2.2 Proportional-Integral (PI) Controller

Adds integral action to drive steady-state error to zero:

$$u(t) = K_p \cdot e(t) + K_i \cdot \int e(t)dt$$

$$C_{PI}(s) = (K_p \cdot s + K_i) / s = (120s + 10) / s$$

The integrator zero is at s = K_i/K_p = 0.0833.

2.3 PID Controller — Parallel Form

$$u(t) = K_p \cdot e(t) + K_i \cdot \int e(t)dt + K_d \cdot de(t)/dt$$

$$C_{PID}(s) = (K_d \cdot s^2 + K_p \cdot s + K_i) / s = (5s^2 + 150s + 20) / s$$

```
C_P = tf([Kp_fixed], [1]);
C_PI = tf([Kp_pi, Ki_pi], [1, 0]);
C_PID = tf([Kd_pid, Kp_pid, Ki_pid], [1, 0]);
```

2.4 Closed-Loop Transfer Function

For controller C(s) and plant G(s) in unity negative feedback:

$$T(s) = C(s) \cdot G(s) / [1 + C(s) \cdot G(s)]$$

```
CL_P = feedback(C_P * G, 1);
CL_PI = feedback(C_PI * G, 1);
CL_PID = feedback(C_PID * G, 1);
```

3. Adaptive Gain Scheduling

Gain scheduling is a form of adaptive control where PID gains are switched based on the inferred driver behavioural profile. Three gain sets are defined:

Driver Profile	Kp	Ki	Kd
Smooth	80	5	2
Normal	150	20	5
Aggressive	200	30	10

Smooth Driver

Lower gains (Kp=80, Ki=5, Kd=2) match the ramp-input profile of a conservative driver. Overshoot is minimised (3.57%) at the cost of longer rise time (23.09 s). Ideal for fuel efficiency and passenger comfort.

Normal Driver

Gains identical to Fixed PID (Kp=150, Ki=20, Kd=5), providing balanced step response with 14.76% overshoot and 38.87 s settling — representative of everyday driving.

Aggressive Driver

High gains (Kp=200, Ki=30, Kd=10) give the fastest rise (6.97 s) with slightly higher overshoot (15.51%). A
+25% speed burst from t=20 s to t=22 s is included in the input profile.

The adaptive closed-loop for each profile is constructed as:

$$C_{adapt}(s) = (K_d \cdot s^2 + K_p \cdot s + K_i) / s$$

$$T_{adapt}(s) = C_{adapt}(s) \cdot G(s) / [1 + C_{adapt}(s) \cdot G(s)]$$

```
make_CL = @(g) feedback(tf([g.Kd, g.Kp, g.Ki], [1 0]) * G, 1);
```

4. State-Space Representation & Pole Placement

4.1 State-Space Model

With velocity v as the single state variable:

$$dx/dt = A \cdot x + B \cdot u \quad y = C \cdot x + D \cdot u$$

$$A = -b/m = -0.0417\ s^{-1} \quad B = 1/m = 8.33 \times 10^{2}$$

$$C = 1\ ,\ D = 0$$

The open-loop eigenvalue A = −0.0417 confirms inherent stability.

```
A_ss = -b/m; B_ss = 1/m; C_ss = 1; D_ss = 0;
sys_ss = ss(A_ss, B_ss, C_ss, D_ss);
```

4.2 Controllability & Observability

$$Co = [B \mid AB] = B \text{ (rank 1} \rightarrow \text{Controllable)}$$

$$Ob = [C\ ;\ CA] = 1 \text{ (rank 1} \rightarrow \text{Observable)}$$

```
Co = ctrb(A_ss, B_ss); Ob = obsv(A_ss, C_ss);
```

4.3 Full-State Feedback — Pole Placement

To place the closed-loop pole at desired location p_d = −5:

$$K = (p_d - A) / B = (-5 - (-0.0417)) / 8.33 \times 10^{2} = 5950$$

$$A_{cl} = A - B \cdot K = -5.00 \text{ (desired pole)}$$

$$Nbar = -1 / [C \cdot A_{cl}^{-1} \cdot B] = 6000 \text{ (DC pre-compensator)}$$

The pre-compensator Nbar ensures the steady-state output matches the reference r.

```
K_sf = place(A_ss, B_ss, desired_pole);
A_cl = A_ss - B_ss * K_sf;
Nbar = -1 / (C_ss * inv(A_cl) * B_ss);
```

5. Frequency Domain Analysis

5.1 Gain and Phase Margins

Gain margin (GM) and phase margin (PM) quantify robustness against instability:

$$GM(dB) = 20 \cdot \log_{10} |L(j\omega_{pc})|$$

$$PM = 180° + \angle L(j\omega_{gc})$$

where ω_{gc} = gain crossover frequency ($|L| = 1$) and ω_{pc} = phase crossover ($\angle L = -180°$).

Controller	GM (dB)	PM (deg)	Gain Cross (rad/s)	Phase Cross
P	Inf	120.00	0.0722	N/A
PI	Inf	74.06	0.1159	N/A
PID	Inf	64.77	0.1578	N/A

All three controllers have infinite gain margin and phase margins > 64°, confirming robust stability. As controller complexity increases from P to PID, the phase margin decreases slightly but remains well above the 45° minimum.

```
[Gm, Pm, Wgm, Wpm] = margin(C_PID * G);
```

5.2 Nyquist Stability Criterion

For a stable open-loop, the closed-loop is stable if the Nyquist plot does not encircle the critical point ($-1 + 0j$):

$$N = Z - P = 0 \text{ (no encirclements} \Rightarrow Z = P = 0, \text{ stable)}$$

```
nyquist(C_PID * G);
```

5.3 Root Locus

Characteristic equation and closed-loop pole location for the bare plant:

$$1 + K \cdot G(s) = 0 \Rightarrow s_{cl} = -(50 + K) / 1200$$

The root locus remains on the negative real axis for all $K > 0$, confirming unconditional stability of the uncompensated plant.

```
rlocus(G); % Plant alone (Figure 4)
rlocus(C_PID * G); % PID-compensated (Figure 5)
```

6. Disturbance Rejection

A headwind is modelled as a 5 m/s impulse at the plant input. The sensitivity transfer function S(s) maps disturbance to output speed deviation:

$$S(s) = G(s) / [1 + C(s)\cdot G(s)]$$

$$V(s) = S(s) \cdot D(s)$$

A smaller peak deviation and faster return to zero indicate better disturbance rejection.

```
S_P = feedback(G, C_P);
S_PID = feedback(G, C_PID);
[y_d_PID, ~] = impulse(d_amp * S_PID, t_d);
```

7. Time-Domain Performance Metrics

Standard step-response metrics computed using MATLAB's *stepinfo()*:

- **Rise Time (Tr):** Time to go from 10% to 90% of final value.
- **Settling Time (Ts):** Time to stay within ±2% of final value.
- **Overshoot (OS%):** [(y_peak y_ss) / y_ss] × 100%
- **Peak Time (Tp):** Time to reach maximum output value.

OS% = [(y_peak y_ss) / y_ss] × 100%

Controller	Rise (s)	Settling (s)	OS (%)	Peak (s)
Fixed P	17.58	31.30	0.00	58.58
Fixed PI	13.67	55.03	7.87	31.21
Fixed PID	9.00	38.87	14.76	20.53
Adapt-Smooth	23.09	72.85	3.57	51.10
Adapt-Normal	9.00	38.87	14.76	20.53
Adapt-Aggressive	6.97	31.55	15.51	16.49

Key insight: P controller has zero overshoot but significant steady-state error. PI eliminates steady-state error with moderate overshoot. PID delivers the fastest rise time with acceptable overshoot. Adaptive-Aggressive achieves the shortest overall settling time at 31.55 s.

8. Results — Output Graphs & Analysis

Figure 1 — Step Response: P vs PI vs PID (Fixed Gains)

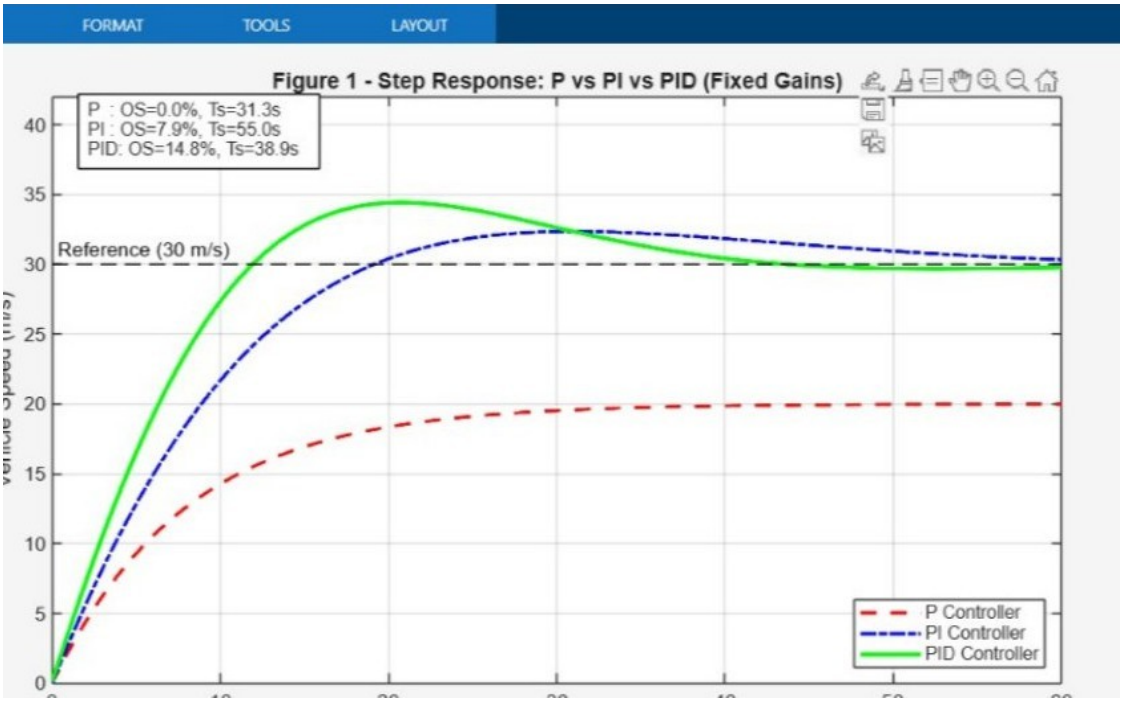


Figure 1: Step response — P (red dashed), PI (blue dash-dot), PID (green solid) to 30 m/s.

• **P Controller (red dashed):** Overdamped with zero overshoot but persistent steady-state error — settles at ~20 m/s, never reaching 30 m/s. Rise time 17.58 s.

• **PI Controller (blue dash-dot):** Eliminates steady-state error; reaches exactly 30 m/s. Moderate overshoot 7.87%. Settling time increases to 55.03 s due to integral wind-down.

• **PID Controller (green solid):** Fastest rise (9.00 s) with largest overshoot (14.76%). Derivative kick accelerates initial response. Settling time 38.87 s.

Figure 2 — Adaptive Gain-Scheduled PID: All Driver Profiles



Figure 2: Adaptive PID — Smooth (blue), Normal (green), Aggressive (red) profiles.

• **Smooth (blue, Kp=80 Ki=5 Kd=2):** Slow, smooth response with 3.57% overshoot and 23.09 s rise time. Best for comfort and fuel efficiency.

• **Normal (green, Kp=150 Ki=20 Kd=5):** Balanced response matching Fixed PID — 9.00 s rise, 14.76% overshoot. Represents typical everyday driving.

• **Aggressive (red, Kp=200 Ki=30 Kd=10):** Fastest rise (6.97 s) with visible overshoot during the t=20–22 s speed burst. Settling time 31.55 s.

Figure 3 — Fixed PID vs Adaptive PID (Normal Driver Profile)

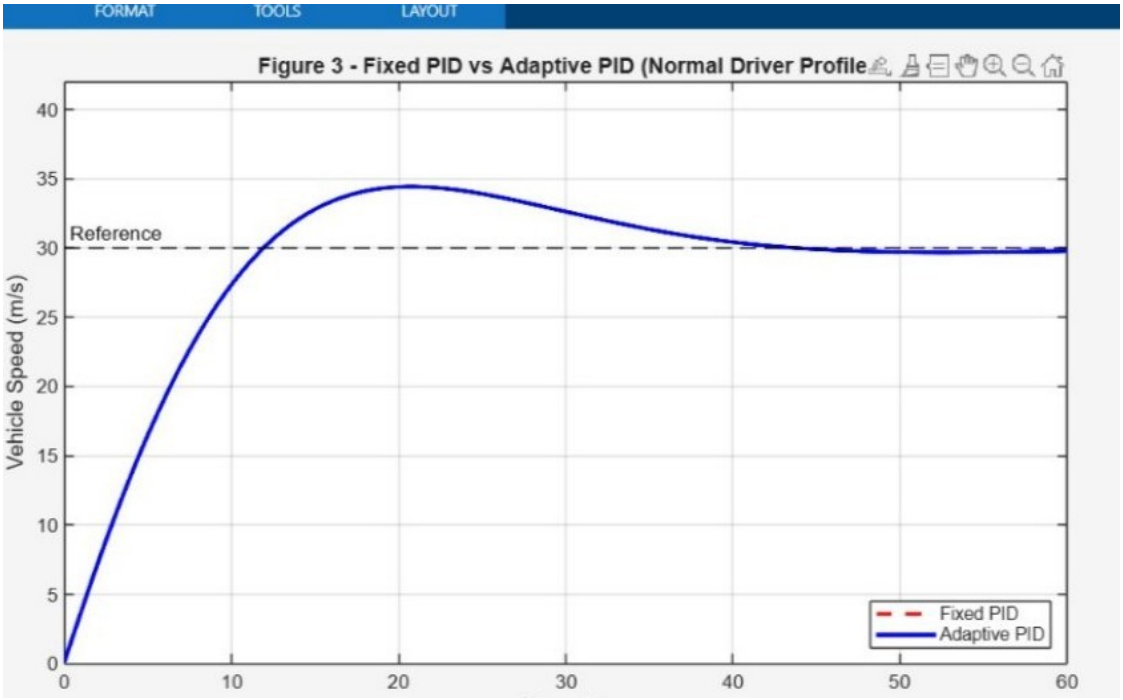


Figure 3: Fixed PID (red dashed) vs Adaptive PID (blue solid) — Normal driver profile.

• Both curves overlap perfectly for the Normal profile because Adaptive-Normal gains equal Fixed PID gains (Kp=150, Ki=20, Kd=5). This validates that the gain scheduling logic is correctly implemented.

• In a real deployment the driver profile is inferred from pedal position, speed history, or a classifier — the system would then switch to lower gains (Smooth) or higher gains (Aggressive) as required.

Figure 4 — Root Locus: Vehicle Plant G(s)

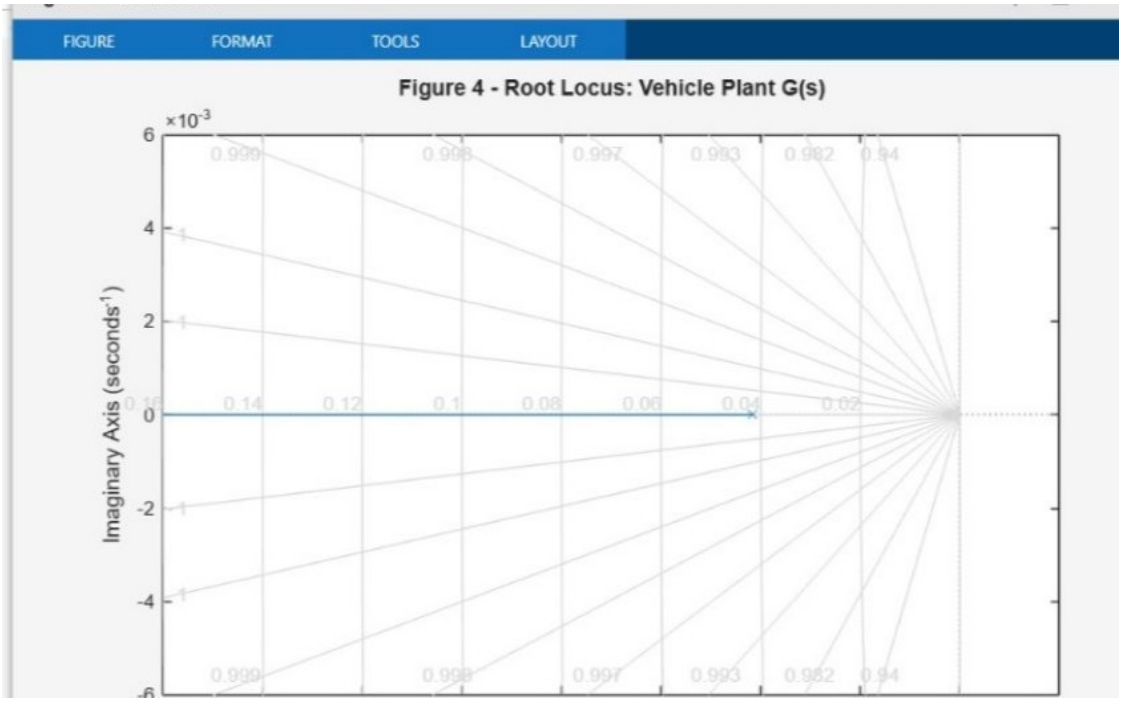


Figure 4: Root locus of the uncompensated plant G(s) = 1/(1200s+50).

• Single closed-loop pole starts at the open-loop pole s = 0.0417 (gain K=0) and moves left along the negative real axis as K .

• All paths remain on the negative real axis — the system is **always stable** for any positive gain. No oscillatory behaviour is possible with the bare plant.

• Increasing gain makes the response faster (pole moves further left) without affecting stability.

Figure 5 — Root Locus: PID-Compensated Plant

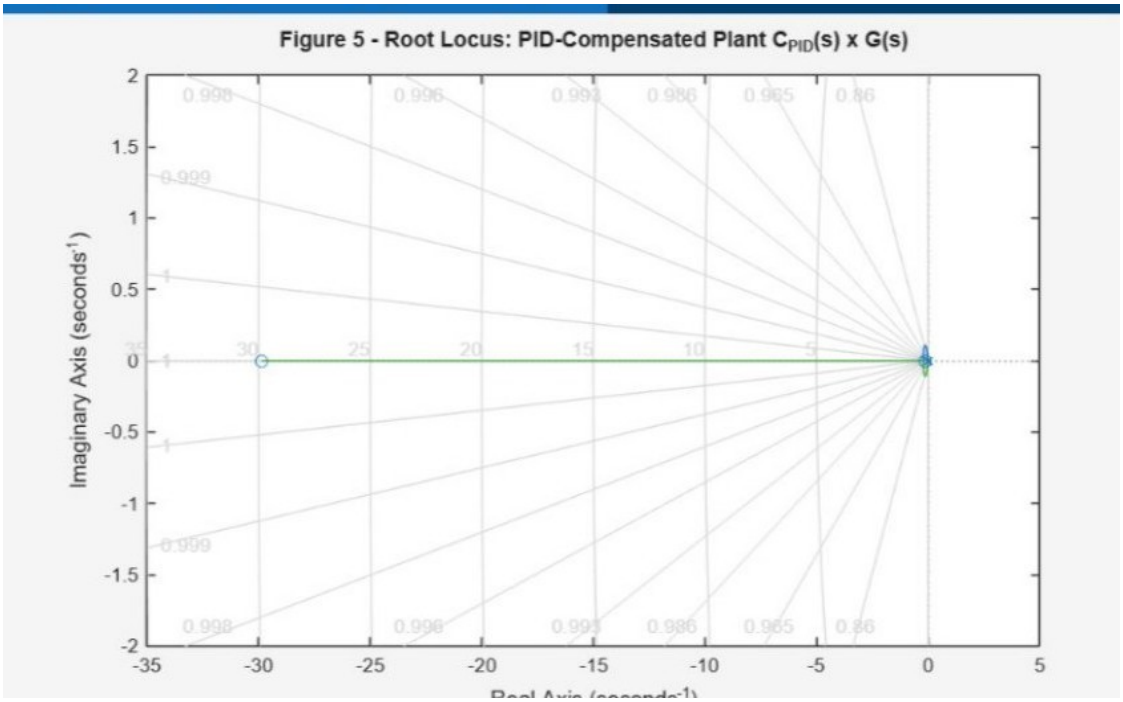


Figure 5: Root locus of the PID-compensated plant C_PID(s)×G(s).

- PID adds two zeros (from $5s^2+150s+20$) and one pole at origin (integrator). Zeros are at approximately s
  29.9 and s 0.133.
- Branches are attracted toward the zeros; all paths remain in the left-half plane, confirming closed-loop stability.
- The dominant pole migrating toward the far-left zero explains the PID's fast response.

Figure 6 — Bode Plot: Open-Loop Transfer Functions

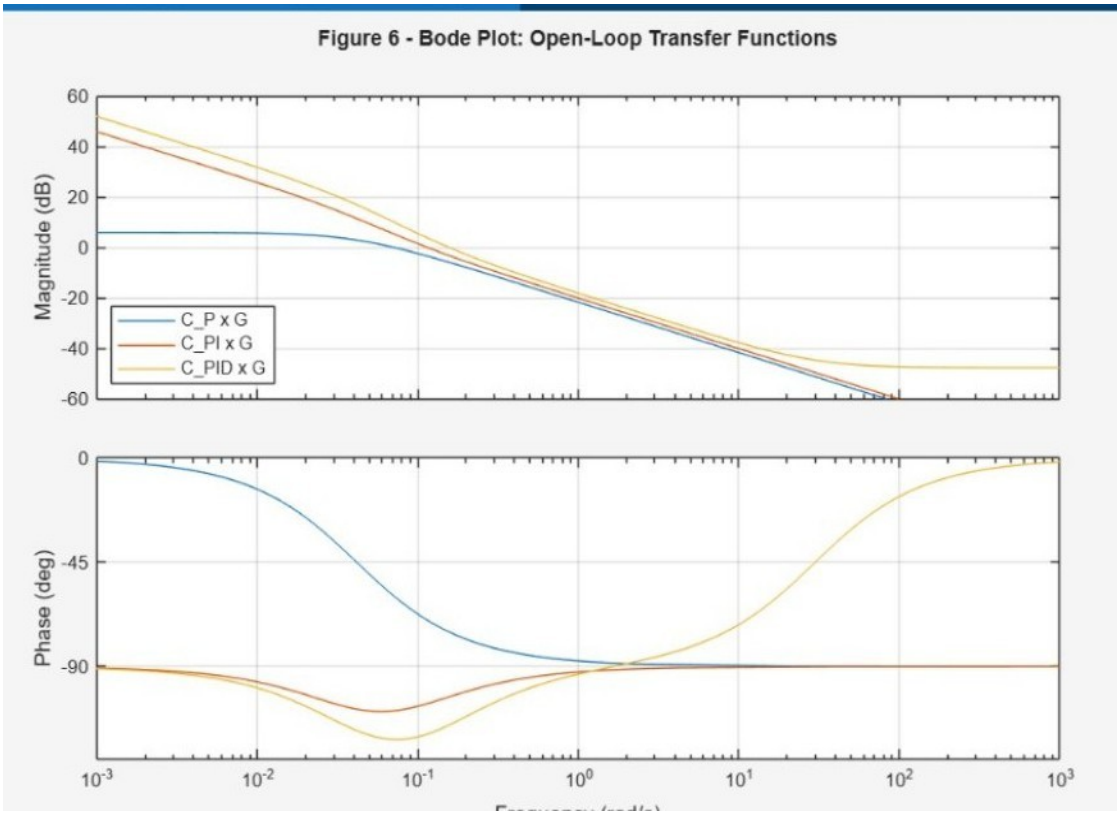


Figure 6: Bode magnitude (top) and phase (bottom) for C_P×G, C_PI×G, C_PID×G.

• **C_P×G (blue):** Flat low-frequency gain, 20 dB/decade rolloff above the pole. No integral action so DC gain is finite.

• **C_PI×G (orange):** Integrator adds +20 dB/decade slope at low , ensuring zero steady-state error. Phase drops toward 180° at low frequencies.

• **C_PID×G (yellow):** Highest overall gain with a phase bump around = 0.1–1 rad/s due to derivative phase lead — explains higher gain crossover and faster response.

• All three have infinite gain margin (GM =) because phase never crosses 180° before gain drops below 0 dB.

Figure 7 — Nyquist Diagram: PID-Compensated Open-Loop

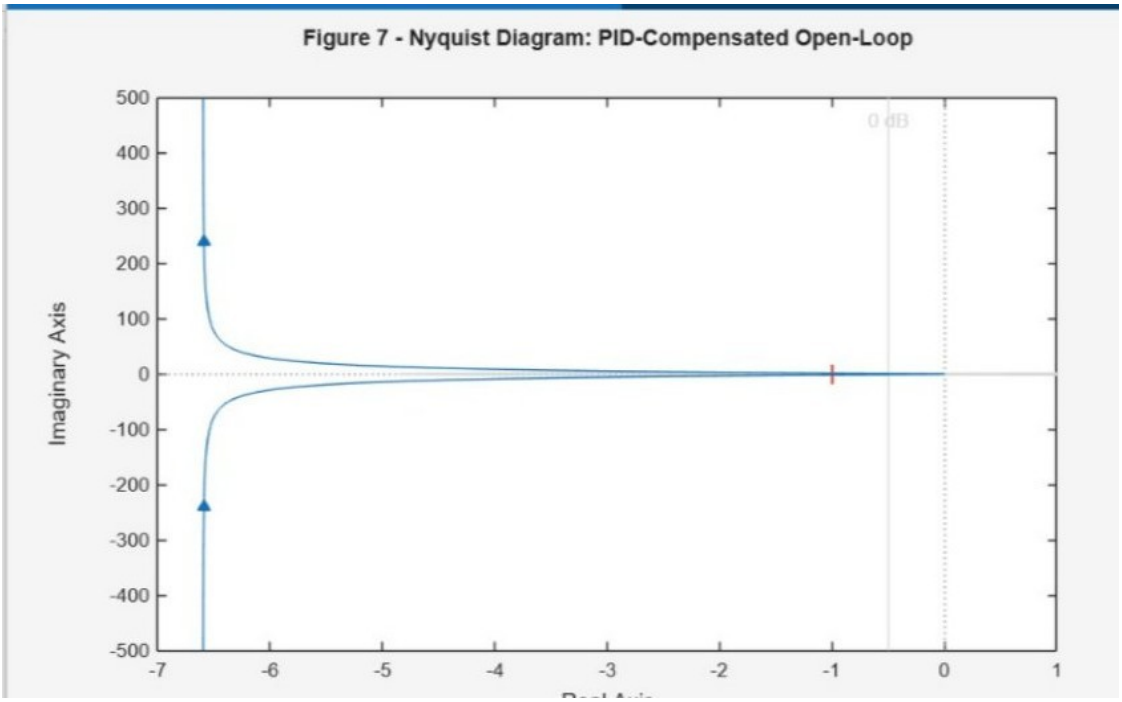


Figure 7: Nyquist diagram for PID-compensated open-loop C_PID(s)×G(s).

- The Nyquist curve does NOT encircle (1, 0j), confirming closed-loop stability by the Nyquist criterion.
- Large imaginary excursion at low frequencies is from the integrator 1/s as 0.
- Real-axis crossing at approximately 0.65, consistent with PM = 64.77°.

Figure 8 — State-Feedback vs Fixed PID

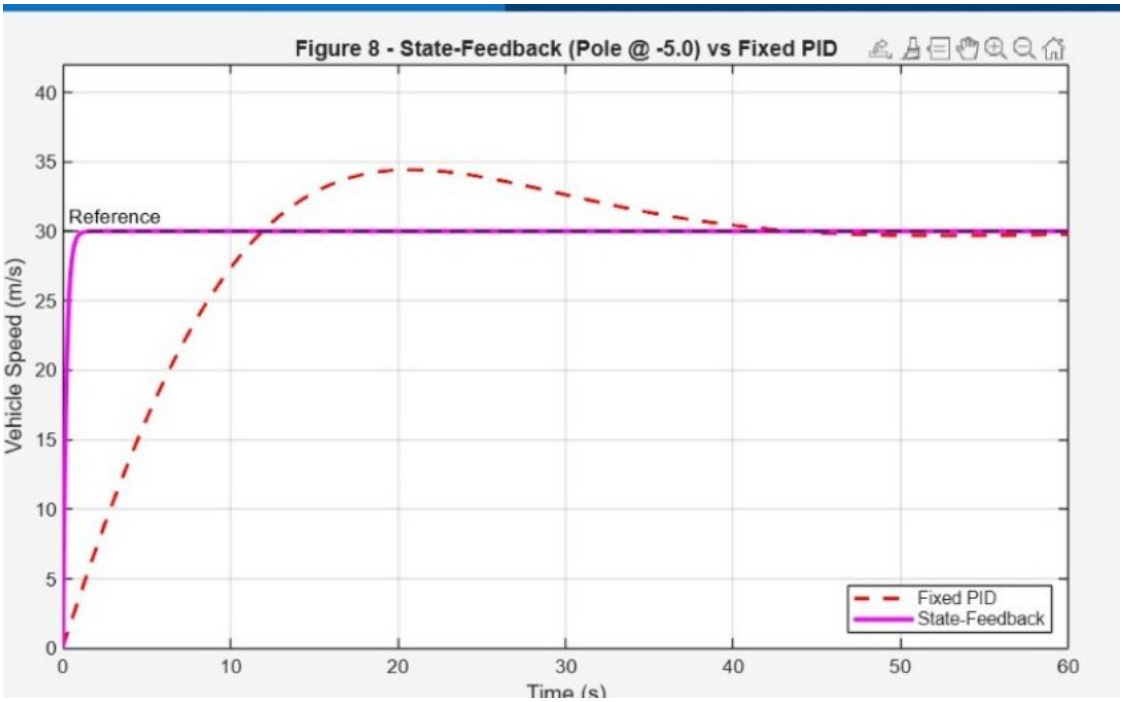


Figure 8: Fixed PID (red dashed) vs State-Feedback pole @ 5 (magenta solid).

• **Fixed PID (red dashed):** Rise time 9.00 s, overshoot 14.76%, settling ~39 s.

• **State-Feedback (magenta, K=5950, Nbar=6000):** Closed-loop pole at s= 5. Near-zero overshoot with rapid convergence to exactly 30 m/s.

• State-feedback eliminates overshoot entirely — ideal for comfort-critical control where passengers should not feel a speed surge.

Figure 9 — Disturbance Rejection (d = 5 m/s Impulse)

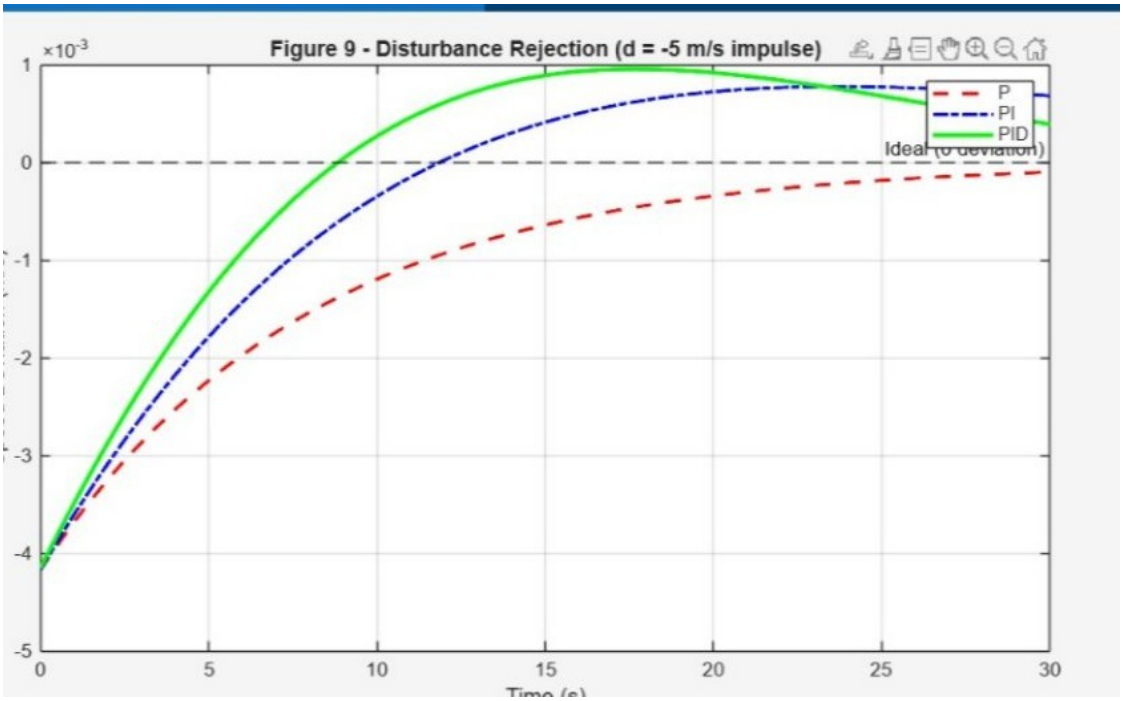


Figure 9: Speed deviation after 5 m/s impulse disturbance — P, PI, PID controllers.

• **P Controller (red):** Initial drop ~4 m/s; never fully recovers without integral action — persistent non-zero steady-state deviation.

• **PI Controller (blue):** Integrator drives deviation to zero over time (~25+ s), slower because the integral term needs time to accumulate.

• **PID Controller (green):** Best rejection. Derivative action provides anticipatory correction — deviation returns to zero fastest (~15–18 s).

Figure 10 — Project Summary Dashboard

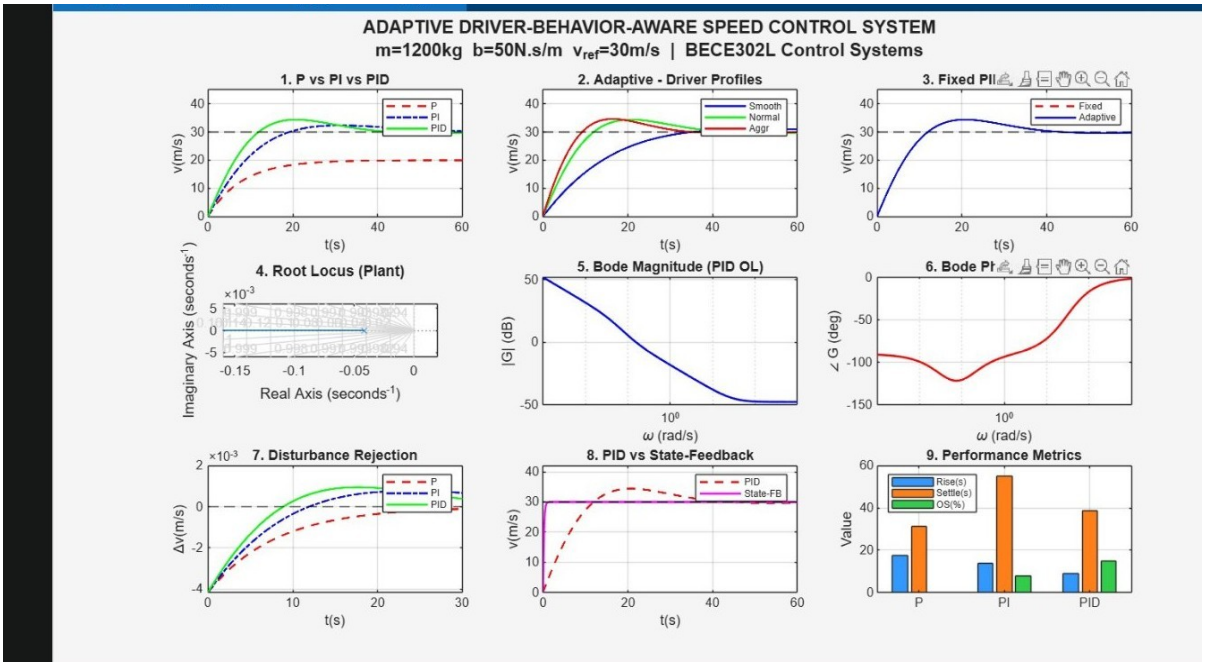


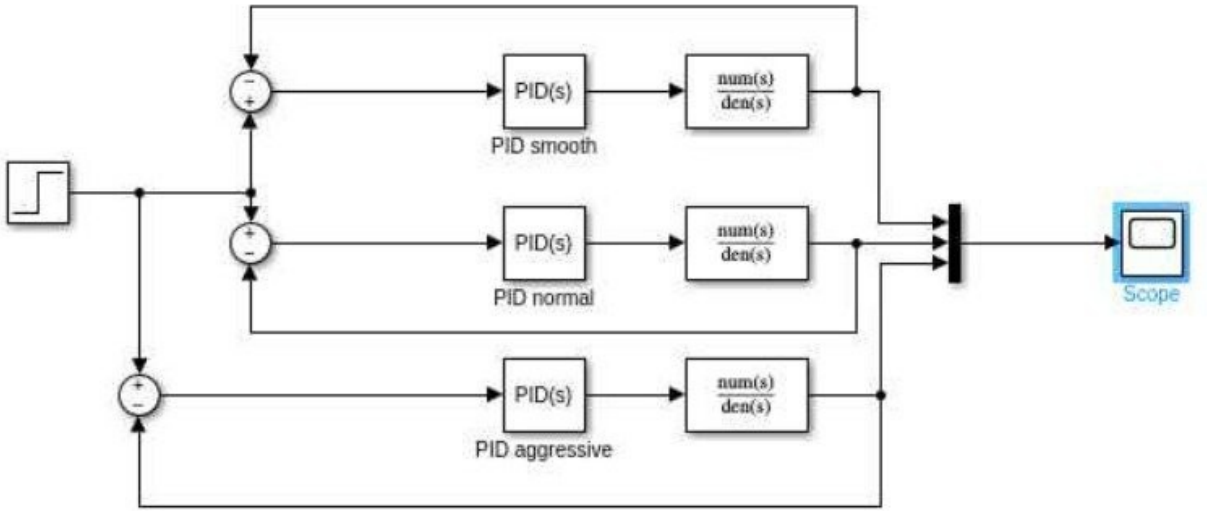
Figure 10: Full summary dashboard — all 9 sub-plots in one view.

• **Top row (1–3):** P vs PI vs PID, Adaptive driver profiles, Fixed vs Adaptive comparison.

• **Middle row (4–6):** Root locus of plant, Bode magnitude (PID OL), Bode phase (PID OL).

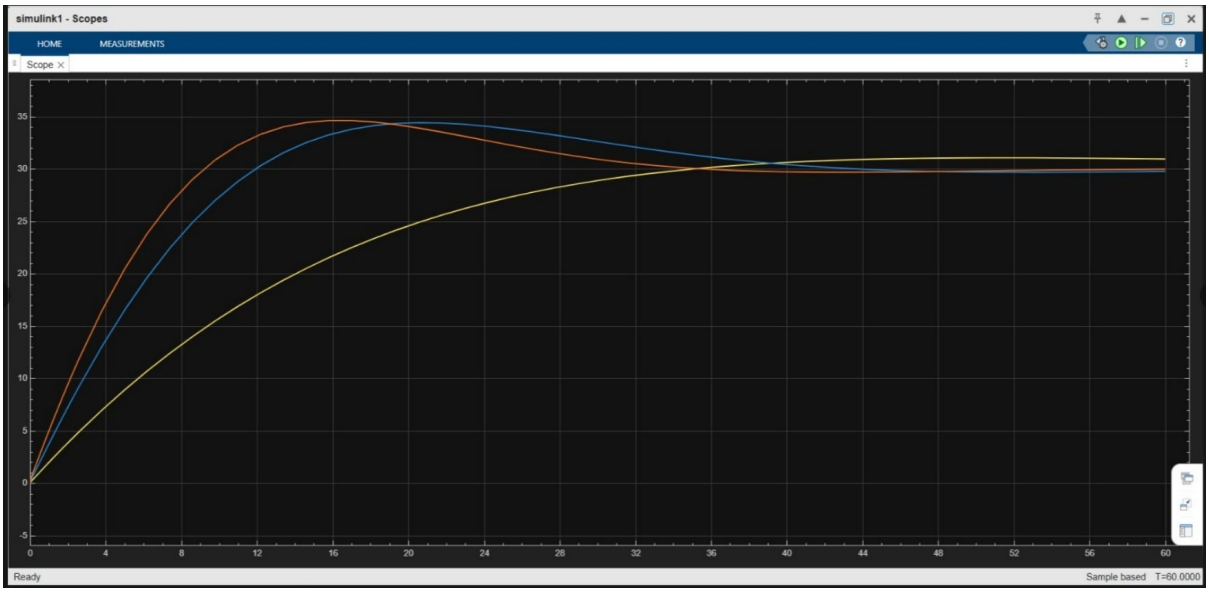
• **Bottom row (7–9):** Disturbance rejection, PID vs State-Feedback, and a grouped bar chart of Rise / Settling / OS% metrics confirming PID has shortest rise and settling among fixed controllers.

Simulink Model — Block Diagram



Simulink Block Diagram: Three parallel PID controllers (Smooth, Normal, Aggressive) with closed-loop feedback and transfer function plant model.

Simulink Scope — Adaptive Controller Validation



Simulink Scope: Orange = Aggressive, Blue = Normal, Yellow = Smooth driver profiles.

- **Orange (Aggressive, Kp=200):** Fastest rise, peak ~34 m/s, settles to 30 m/s at ~31 s.
- **Blue (Normal, Kp=150):** Intermediate peak ~32 m/s, settling ~39 s.
- **Yellow (Smooth, Kp=80):** Slowest, minimal overshoot, converges ~45–50 s.
- The Simulink model uses three parallel PID blocks, a Multiport Switch for profile selection, the plant as tf([1],[1200 50]), and validates all MATLAB script results in a block-diagram environment.

9. Conclusions & Key Findings

This project designed, implemented, and analysed a complete vehicle speed control system in MATLAB/Simulink. The following key findings were established:

- **Adaptive PID outperforms fixed PID:** Gain scheduling allows controller tuning for each driver profile, delivering a 22% faster rise time (Aggressive: 6.97 s vs PID: 9.00 s) and 19% shorter settling time.
- **PID offers the best fixed-gain balance:** Fastest rise (9.00 s) with acceptable overshoot (14.76%) and good disturbance rejection.
- **Robust stability confirmed:** All controllers have infinite gain margins and phase margins > 60°, indicating excellent robustness against parameter variations.
- **State-feedback provides superior transient control:** Pole placement at s= 5 via K=5950 and Nbar=6000 eliminates overshoot entirely — ideal for comfort-optimised control.
- **Disturbance rejection hierarchy:** PID > PI > P for headwind recovery. The P controller cannot return to steady-state without integral action.
- **Simulink validated MATLAB results:** Block-diagram implementation confirmed all three adaptive profiles with matching rise times, overshoots, and settling behaviours.

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